MILITARY SPECIFICATION SHEET

ELECTRON TUBE, CATHODE RAY

TYPE 10ANP7

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of Specification MIL-E-1.

DESCRIPTION: 10 inch, magnetic deflection, magnetic focus cathode ray tube with aluminized screen

DIMENSIONS AND PIN CONNECTIONS: See figure 1

ABSOLUTE RATINGS:

Parameter:	Ef	Ec1	Ec2	Eb	Ehk	Rg1	Alt
Unit:	V	Vdc	Vdc	Vdc	Vdc	Meg	ft
Maximum:	6.9	0	450	11,000	+125	1.5	20,000
Minimum:	5.7	-200	---	7,000	-125	---	---
TEST CONDITIONS:	6.3	Adj (Note 1)	250	9,000	---	---	---

GENERAL:

Qualification - Required (see note 5)

Ⓔ denotes changes

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METHOD	REQUIREMENT OR TEST	NOTES	CONDITIONS	SYMBOL	LIMITS		UNIT
					MINIMUM	MAXIMUM	
Ⓔ	Qualification inspection						
5216	Cathode illumination	-		---	---	---	---
1331	Direct-interelectrode capacitance	-	k to all g1 to all	Ck Cg1	--- ---	5.0 6.3	pF pF
5201	Electrode current (grid No. 2)	-	Ec1 = 0	Ic2	---	5.0	µAdc
1141	Pressure (implosion)	-		---	---	---	---
1002	Barometric pressure, reduced	7		---	---	---	---
5111	Vibration	-		Width	---	2.0	mm
	Quality conformance inspection, part 1						
5201	Voltage breakdown	-		---	---	---	---
5201	Voltage breakdown (magnetic types)	-		---	---	---	---
5206	Gas ratio	6		Gr	---	0.25	---
5101	Neck straightness	2		---	---	---	---
5106	Screen and faceplate blemishes	-		---	---	---	---
5223	Modulation	1	Ib = 400 µAdc	ΔEc1	---	40	Vdc
5231	Spot position (magnetic deflection)	-		Radius	---	12.5	mm
5241	Grid cutoff voltage	-		Ec1	20	45	Vdc
5246	Focus coil current	-	Ib = 100 µAdc; D = 2.5 inch using JEDEC No. 127 focus coil, or equivalent	---	40	60	mAdc
---	Aperture alinement	4		---	---	8	mm
5251	Grid No. 1 leakage current	-		---	---	---	---
	Quality conformance inspection, part 2						
1301	Heater current	-		If	270	330	mA
5101	Neck and bulb (magnetic types)	-		---	---	---	---
5101	Side terminal and base alinement	-	Halfway between 1 and 9	---	-10	+10	Degrees
5101	Face tilt	-		---	---	---	---
5216	Stray light emission (conventional types)	-	Eb = 11,000 Vdc; Ec2 = 450 Vdc	---	---	---	---
5221	Screens	3		---	---	---	---
5226	Line width A (magnetic deflection)	-	Ib = 100 µAdc	Width	---	0.33	mm
5226	Line width C (magnetic deflection)	-	Ib = 100 µAdc	Width	---	0.50	mm
5251	Heater-cathode leakage current	-		---	---	---	---
5251	Grid No. 2 leakage current	-		---	---	---	---

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METHOD	REQUIREMENT OR TEST	NOTES	CONDITIONS	SYMBOL	LIMITS		UNIT
					MINIMUM	MAXIMUM	
	Quality conformance inspection, part 2 -Continued						
Ⓔ 1105	Permanence of marking	-		---	---	---	---
	Quality conformance inspection, part 3						
---	Life test	-	Group C; Eb = 11,000 Vdc; Ec2 = 450 Vdc; Ib = 60 µAdc	t	500	---	hrs
---	Life-test end points:						
5226	Line width A	-	Ib = 60 µA	Width	---	0.33	mm
5223	Modulation	1	Ib = 320 µA	ΔEc1	---	40	Vdc

NOTES:

1. ΔEc1 shall be not greater than the value of grid cutoff voltage (see method 5241).
2. Cylinder shall be 4 inches long and 0.910 inch maximum inside diameter.
3. The screen characteristics shall be measured with constant beam energy of 0.24 watt defocused to a spot approximately 0.25 cm in diameter. The test conditions shall be anode voltage (relative to cathode) 9,000 volts and beam current 27 µAdc maximum. The screen characteristics shall comply with the following minimum limits: cB5 = 480 cB; G5: 1 = 4.
4. The distance between the center of the unfocused, undeflected spot at low intensity (Ec1 near cutoff) and the center of the image of the masking aperture observed at high intensity of the unfocused, undeflected spot shall not exceed the limit specified herein. Ec1 shall not be held at zero for more than approximately 10 seconds to prevent damage to the screen.
5. The activity responsible for the qualified products list is the Naval Electronic Systems Command, Department of the Navy, Washington, D.C. 20360, and information pertaining to qualification of products may be obtained from that activity. Application for Qualification tests shall be made in accordance with "Provisions Governing Qualification SD-6". (Copies of "Provisions Governing Qualification SD-6" may be obtained upon application to Commanding Officer, Naval Publications and Forms Center, 5801 Tabor Avenue, Philadelphia, Pennsylvania 19120.)
6. This test is to be performed at the conclusion of the holding period.
7. The low-pressure test shall be made with maximum voltages applied, and pressure equivalent to 20,000 feet (348 mmHg). Connections shall be made to the base pins in a manner that does not degrade the tube breakdown characteristics. Satisfactory operation shall be the absence of arc-over and harmful corona.

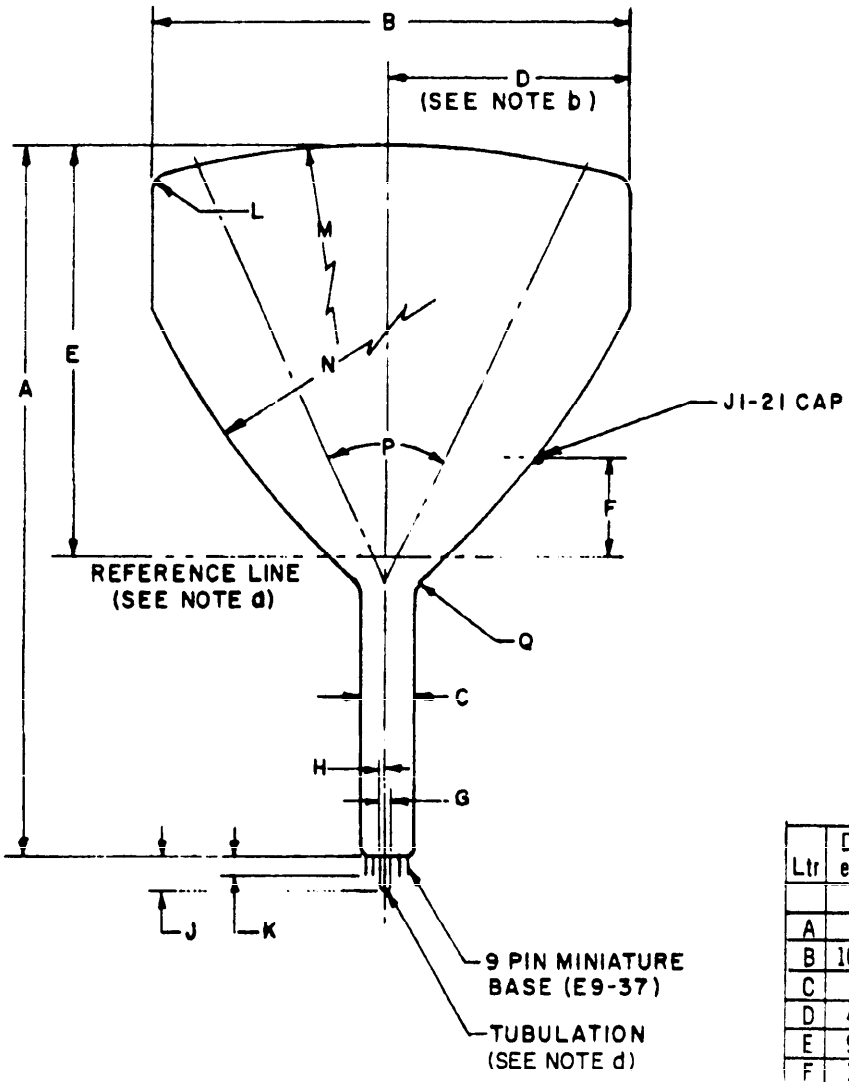
User activities:
Navy - AS, OS, MC, CG, SH

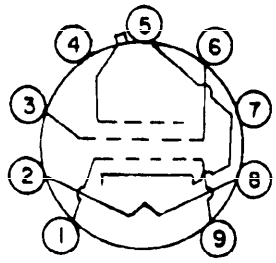
Preparing activity:
Navy - EC

Agent:
DSA - ES

(Project 5960-N066)

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BOTTOM VIEW

Pin No.	Element
1 —	Grid No. 1
2 —	Heater
3 —	Grid No. 2
5 —	Cathode
6 —	Grid No. 2
8 —	Heater
9 —	Grid No. 1
Cap —	Accelerator

Ltr	Dimensions in inches with metric equivalents (mm) in parentheses	
	Minimum	Maximum
A		16.00 (406.40)
B	10.437 (265.10)	10.563 (268.30)
C	.81 (20.57)	.896 (22.76)
D	4.50 (114.30)	
E	9.233 (234.52)	9.483 (240.87)
F	1.915 (48.64)	2.415 (61.34)
G		.250 (6.35)
H		.125 (3.18)
J		.437 (11.10)
K		.250 (6.35)
L	.437 (11.10) Rad	
M	42.00 (1066.80) Rad	
N	20.00 (508.00) Rad	
P	50°	
Q	.500 (12.70) Rad	

NOTES:
a. Reference line is determined by the point where JEDEC G-141 reference line gage will stop.
b. Useful screen radius.
c. Dimensions without tolerances are for information only.
d. The tubulation diameter shall be less than the diameter of the socket hole and the tubulation shall not cant so as to bear on the sides of the socket hole.

FIGURE 1. Outline drawing of electron tube type 10ANP7.

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

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REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

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